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Artificial intelligence in food demand forecasting

Artificial Intelligence and Machine Learning for Food Demand Forecasting in Institutional Food Services: Methods, Outcomes, and Research Gaps

Quick Reference

Key Findings Table

Theme	Summary	Supporting Citations
Diverse Predictive Techniques	Ensemble trees, neural networks, and hybrid models are most effective; hybrid models balance accuracy and interpretability.	1 2 3
IoT & Big Data Integration	Real-time sensor data and big data analytics significantly improve forecasting accuracy and operational responsiveness.	4 5 6
Explainability & Trust	Interpretable models and XAI tools (SHAP, LIME, E2Tree) are crucial for stakeholder trust and operational adoption.	7 8 9
Operational Impact Metrics	Waste reduction, stock turnover, and sustainability KPIs are increasingly used alongside accuracy metrics.	10 11 12
Barriers & Research Gaps	Data quality, infrastructural challenges, and lack of standardized datasets limit scalability and generalizability.	13 14 15

Direct Answer

Artificial intelligence (AI) and machine learning (ML) can be used to predict food demand in institutional food services by leveraging diverse modeling techniques such as nonlinear autoregressive neural networks, ensemble methods, and hybrid models that combine the strengths of decision trees and neural networks. These methods are often integrated with IoT sensors and big data analytics to provide real-time forecasting, enabling food services to minimize food waste, optimize food preparation, and improve inventory management. Key findings include high prediction accuracies (with R2 values often exceeding 96%), improved waste reduction, and operational efficiencies. However, challenges such as data heterogeneity, model interpretability, infrastructural barriers, and the need for open-access, standardized datasets remain significant limitations. Emerging trends suggest a shift towards interpretable hybrid models, integration with IoT and blockchain technologies, and the use of advanced explainability techniques to enhance trust and usability.
1 4 5 7 10 13 14 15 16

Study Scope

- **Time Period:** 2015–2024

- **Disciplines:** Food service operations, supply chain management, AI/ML, sustainability, IoT, blockchain
- **Methods:** Empirical studies, meta-analyses, case studies, algorithmic benchmarking, operational impact assessments

Assumptions & Limitations

- Most studies rely on proprietary or limited datasets, restricting generalizability.
- Operational impact metrics are not standardized across studies.
- Deployment challenges (cost, infrastructure, workforce skills) are more pronounced in low- and middle-income countries.
- Explainability and trust are critical but under-researched in practical deployments.

Suggested Further Research

- Development of standardized, open-access datasets and validation frameworks.
- Exploration of dynamic, real-time adaptive AI systems integrating external factors (weather, holidays, socio-economic indicators).
- Integrated decision support platforms combining AI predictions, sustainability metrics, and real-time inventory status.
- Context-sensitive strategies for AI adoption in low-resource settings.

1. Introduction

Background and Motivation

Food waste in institutional settings—such as universities, hospitals, and military facilities—remains a significant operational and sustainability challenge. Globally, institutional food services contribute substantially to food waste, with estimates ranging from 14% to 30% of total food loss in developed economies. This waste not only represents lost resources but also exacerbates environmental impacts and operational inefficiencies. Traditional forecasting methods often fail to capture the complex, dynamic nature of food demand, leading to overproduction, spoilage, and inventory mismanagement.
AI and ML offer transformative potential by enabling more accurate, real-time demand forecasting, thus supporting waste minimization, optimized food preparation, and improved inventory management. The integration of these technologies with IoT sensors and big data analytics further enhances their effectiveness, providing actionable insights for institutional food service managers and supporting sustainability goals [16] [17] [18].

2. Theoretical Frameworks

Commonly Used AI and ML Techniques

- **Neural Networks:** Nonlinear autoregressive exogenous neural networks achieve high accuracy ($R2 > 96\%$) for processed foods, supporting inventory and production control [16].
- **Ensemble Decision Trees:** Random Forest, Gradient Boosting, and Extra Trees are robust and interpretable, often outperforming traditional statistical methods [1] [19] [20].
- **Hybrid Models:** Combine neural networks and decision trees to balance accuracy and interpretability, especially in complex institutional contexts [1] [2] [3].
- **Regression Techniques:** Ridge Regression, Support Vector Machines, and Partial Least Squares are used for demand prediction and quality assessment [21].
- **Advanced Methods:** XGBoost and CatBoost are applied for food wastage prediction and demand forecasting, showing high predictive accuracy [20] [22].

Synthesis

The diversity of AI/ML techniques reflects the complexity of institutional food demand forecasting. Ensemble tree methods are favored for their interpretability and robustness, while neural networks excel in capturing nonlinear patterns. Hybrid models are increasingly adopted to address the trade-off between accuracy and operational transparency [1] [2] [3].

Comparative Analysis: Ensemble Decision Trees vs. Neural Networks

Model Type	Accuracy	Interpretability	Resource Requirements	Suitability for Institutional Contexts
Ensemble Decision Trees	High	Moderate-High	Moderate	High
Neural Networks	Very High	Low	High	Moderate (resource-dependent)
Hybrid Models	Very High	Moderate	Moderate-High	High

- Ensemble trees (Random Forest, Extra Trees) provide robust accuracy and are easier to interpret, critical for operational decision-making [23] [24].
- Neural networks offer superior accuracy but require more computational resources and are less interpretable [25] [26].
- Hybrid models (e.g., XGBoost + LSTM + SHAP) achieve both high accuracy and interpretability, supporting early intervention and practical deployment [27] [28].

Synthesis

While neural networks can outperform decision trees in accuracy, ensemble trees and hybrid models offer a favorable balance for institutional food services, where interpretability and operational transparency are essential [2] [29].

Integration with IoT and Big Data Analytics

- **IoT Sensors:** Provide real-time environmental data (temperature, humidity, gas composition) for timely detection of anomalies and hazards [30].
- **Big Data Analytics:** Capture spatial and temporal variability, enhancing yield and demand forecasting [4] [31].
- **Edge Computing:** Facilitates low-latency, real-time forecasting and inventory optimization [32].
- **Integration:** Supports predictive analytics for environmental and system condition forecasting, enabling real-time adjustments in food production and supply processes [33] [34].

Synthesis

IoT and big data integration are pivotal for real-time, dynamic forecasting in institutional food services, enabling proactive inventory management and waste reduction [4] [5] [6].

Advances in Model Interpretability and Hybrid Approaches

- **E2Tree:** Transforms ensemble decision paths into a single explainable tree, preserving accuracy and providing global interpretability [7].
- **SHAP & LIME:** Widely used for post hoc explainability, identifying key variables influencing predictions [7] [9].
- **Hybrid Models:** Combine deep learning (CNN-LSTM) with explainability tools for high accuracy and transparency [28].
- **Human-in-the-Loop:** Empowers managers to supervise AI decisions, improving trust and usability [35].

Synthesis

Recent advances focus on balancing predictive accuracy with interpretability, employing explainability tools and hybrid architectures to support decision-making in institutional food demand forecasting [7] [8] [28].

Deployment Challenges and Resource Requirements

- **Computational Resources:** Neural networks require significant resources; ensemble trees are more efficient [22].
- **Infrastructure:** Robust IT infrastructure and data management capabilities are critical [36] [37].
- **Organizational Readiness:** Low industrial maturity and integrability limit practical deployment [36].

- **Embedded Hardware:** Model quantization and pruning enable deployment in resource-limited settings [38].
- **Scalability & Sustainability:** Energy consumption and cost pose challenges for deep learning models [38] [39].

Synthesis

Deployment of AI/ML models in institutional food services is constrained by computational, infrastructural, and organizational factors, necessitating efficient model design and hardware optimization [22] [36] [38].

3. Methods & Data Transparency

- **Data Sources:** Attendance records, menu schedules, consumption patterns, environmental sensor data, historical sales, external factors (weather, holidays, events) [40] [41].
- **Model Training:** Supervised learning with cross-validation, feature engineering to incorporate external variables, ensemble and hybrid model architectures [1] [42].
- **Validation:** Use of accuracy metrics (R2, MAE, RMSE), operational impact metrics (waste reduction, stock turnover), and explainability tools (SHAP, LIME) [7] [10].
- **Transparency:** Interactive dashboards and user interfaces facilitate data sharing, monitoring, and actionable insights for stakeholders [43] [44].

Synthesis

Transparent data practices and robust validation frameworks are essential for reliable AI/ML-based food demand forecasting in institutional settings. Interactive dashboards enhance usability and adoption [43] [44].

4. Critical Analysis of Findings

Key Findings and Performance Results

- **Accuracy:** Ensemble and hybrid models achieve high prediction accuracy ($R2 > 96\%$), outperforming traditional methods [19] [22].
- **Waste Reduction:** AI-driven systems reduce food waste by up to 37% and per-meal carbon footprint by 22% in institutional settings [6].
- **Inventory Optimization:** Improved stock turnover ratios, fill rates, and supply chain efficiency metrics [10] [45].
- **Cost Savings:** AI-based models can reduce total life cycle costs by up to 18% compared to conventional methods [12].
- **Contextual Performance:** Models perform well across hospitals, universities, and restaurants, but generalizability is limited by data heterogeneity and lack of standardized datasets [22] [46].

Synthesis

AI/ML models deliver substantial operational benefits in institutional food services, including waste reduction, cost savings, and improved inventory management. However, performance varies by context and is limited by data quality and generalizability [6] [22].

Evaluation Metrics and Benchmarks

- **Accuracy Metrics:** R2, MAE, RMSE, MAPE
- **Operational Metrics:** Waste-to-sales ratio, stock turnover, fill rate, cost savings, supply chain efficiency [10] [45].
- **Sustainability Metrics:** Food waste reduction percentage, carbon footprint per meal [6].
- **Benchmarking:** Few standardized frameworks exist; most studies focus on mathematical error metrics [47] [48].

Synthesis

A shift towards comprehensive operational and sustainability metrics is evident, but standardized benchmarking frameworks are lacking [6] [10].

Influence of External Factors

- **Weather:** Temperature and rainfall improve forecast accuracy by up to 19% [41].
- **Holidays & Events:** Proper modeling of calendar effects enhances prediction performance, especially during peak periods [49] [50].
- **Feature Engineering:** Incorporating external variables is essential for improved accuracy and inventory management [42] [51].

Synthesis

External factors significantly impact demand forecasting accuracy, necessitating advanced feature engineering and context-specific modeling [41] [42].

Cost-Benefit Analyses: AI vs. Traditional Methods

- **Traditional Methods:** Multiple regression is accurate but naive methods are preferred for ease of use [52].
- **AI Models:** Offer improved accuracy and cost savings but require higher implementation resources [12].
- **Hybrid Approaches:** Leverage strengths of both for optimal performance [53] [54].

Synthesis

AI models provide superior accuracy and operational benefits, but traditional methods remain competitive due to lower resource requirements. Hybrid approaches are recommended [12] [53].

Operational Impact Metrics Beyond Accuracy

- **Waste-to-Sales Ratio:** Reflects sustainability improvements [10].
- **Stock Turnover & Fill Rate:** Indicate inventory optimization [45].
- **Supply Chain Efficiency:** Enhanced by real-time monitoring and predictive analytics [55] [56].

Synthesis

Operational metrics beyond accuracy are critical for evaluating the real-world impact of AI/ML systems in institutional food services [10] [45].

Contextual Performance Across Institutions and Regions

- **Hospitals, Universities, Military:** AI models improve accuracy and efficiency but face challenges in generalization [22] [46].
- **Geographic Variability:** Data heterogeneity and weak model transferability necessitate context-specific implementations [46] [57].
- **Organizational Factors:** Data governance, workforce skills, and change management influence adoption [47] [58].

Synthesis

Performance varies across institutional contexts and regions, highlighting the need for tailored AI solutions and robust data integration strategies [22] [46].

Sustainability and Circular Economy Benchmarks

- **Food Waste Reduction:** AI-driven planning reduces forecasting errors and waste by 20–40% [47].
- **Carbon Footprint:** IoT-integrated AI systems reduce per-meal carbon footprint by 22% [6].
- **Benchmarking Gaps:** Lack of standardized frameworks for sustainability outcomes [47] [48].

Synthesis

AI/ML systems contribute to sustainability goals, but standardized benchmarking for sustainability outcomes is needed [6] [47].

5. Real-world Implications

AI-Driven Waste Minimization and Inventory Optimization

AI Approaches for Waste Minimization and Inventory Control

- **Predictive Analytics:** ML models (XGBoost, LSTM, Random Forest) optimize demand forecasting and inventory control, minimizing waste and maximizing cost savings [10].
- **Supply Chain Optimization:** Real-time tracking and predictive forecasting reduce spoilage and improve stock control [59] [60].
- **Integration with IoT & Blockchain:** Enhances transparency, traceability, and real-time monitoring [55] [56].

Synthesis

AI/ML approaches significantly reduce food waste and optimize inventory management in institutional food services, supporting sustainability and operational efficiency [10] [59] [60].

Computer Vision and Real-Time Waste Quantification

- **SmartMessAI:** Integrates attendance, menu, and consumption data to predict demand and calculate waste in real-time, reducing food waste in institutional kitchens [40].
- **IoT Sensors:** Combined with AI models (YOLOv5-nano, LSTM, decision trees) monitor food distribution, predict demand, and track carbon footprint [6].
- **Interactive Dashboards:** Provide actionable insights for kitchen staff and administrators [6] [40].

Synthesis

AI-driven computer vision and IoT sensors enable real-time waste quantification and classification, improving waste analytics and management efficiency [6] [40].

AI-Enabled Shelf-Life Prediction and Intelligent Packaging

- **Shelf-Life Prediction:** AI models analyze degradation patterns and real-time environmental data, optimizing production schedules and reducing overproduction [61].
- **Intelligent Packaging:** Sensors and AI algorithms provide real-time monitoring, achieving up to 35% improvement in shelf life and 94.8% spoilage prediction accuracy [62].
- **Case Studies:** AI-powered monitoring systems reduce food waste by 14.8% per store and 30% in kitchens [63].

Synthesis

AI-enabled shelf-life prediction and intelligent packaging significantly reduce overproduction and waste in institutional food supply chains [61] [62] [63].

Blockchain Integration for Transparency and Traceability

- **Blockchain + AI:** Enhances transparency, accountability, and real-time visibility, reducing errors and inefficiencies in inventory management [64] [65].
- **Real-Time Tracking:** Facilitates inventory prediction and waste reduction by leveraging IoT sensor data and smart contracts [66].
- **Challenges:** High implementation costs, scalability, interoperability, and regulatory compliance [66] [67].

Synthesis

Blockchain integration with AI improves transparency and traceability, optimizing inventory management and reducing spoilage in institutional food services [64] [66].

Integrated AI and IoT Systems for Real-Time Decision Support

- **Smart Kitchens:** Real-time monitoring of food freshness, expiry dates, and ingredient availability supports informed decision-making and waste reduction [5].
- **Computer Vision:** Detects and quantifies food waste, providing immediate analytics and dynamic dashboards [17].
- **Challenges:** High costs, workforce adaptation, data privacy, and regulatory frameworks [68] [69].

Synthesis

Integrated AI and IoT systems enable real-time decision support, improving operational efficiency and supporting sustainability goals in institutional kitchens [5] [17].

Role of Interactive Dashboards and User Interfaces

- **Data Sharing & Monitoring:** Enhance usability and accelerate data literacy among stakeholders [43].
- **User-Friendly Interfaces:** Foster trust and confidence in AI technologies, critical for adoption [70] [71].
- **Co-Creation:** Participatory design ensures dashboards are functional and contextually relevant [72].

Synthesis

Interactive dashboards and user interfaces are essential for facilitating adoption and effective use of AI-based food demand forecasting in institutional food services [43] [44].

6. Future Research Directions

Current Research Gaps and Limitations

- **Data Quality & Availability:** Limited datasets and data variability hinder robust predictions [13] [22] [73].
- **Model Generalizability & Interpretability:** Challenges in transferring models across contexts and ensuring transparency [13] [22] [73].

- **Infrastructural Barriers:** High implementation costs and low industrial maturity limit adoption [14] [55].
- **Ethical Concerns:** Transparency, algorithmic bias, and equity issues require governance frameworks [18] [74].
- **Standardization:** Lack of open-access datasets and validation frameworks impedes reproducibility [18] [55] [73].

Synthesis

Addressing data quality, model generalizability, infrastructural barriers, and ethical concerns is critical for advancing AI/ML-based food demand forecasting in institutional settings [13] [14] [18] [73].

Improving Generalizability and Interpretability

- **Ensemble & Hybrid Models:** Improve forecasting performance and generalizability [75].
- **Explainable AI:** SHAP, LIME, and human-in-the-loop approaches enhance transparency and trust [8] [35] [76].
- **Context-Sensitive Design:** Participatory governance and localized adaptation are essential [77] [78].

Synthesis

Strategies to improve generalizability and interpretability include ensemble and hybrid models, explainable AI, and context-sensitive design [8] [75] [76].

Barriers to Adoption in Low- and Middle-Income Countries

- **Infrastructural Gaps:** Limited digital infrastructure and real-time data deficits [15] [79].
- **Socio-Economic Barriers:** Skill shortages, workforce readiness gaps, and financial constraints [15] [47] [80].
- **Regulatory & Governance:** Lack of national AI strategies and standards for explainability and accountability [77] [81].
- **Behavioral Resistance:** Organizational rigidity and low awareness require foundational investments [15] [82].

Synthesis

AI adoption in institutional food services in LMICs is constrained by interdependent infrastructural, socio-economic, ethical, and organizational barriers, requiring sequenced interventions [15] [47].

Trust, Usability, and Explainable AI

- **XAI Techniques:** SHAP, LIME, and PDP enhance transparency and interpretability [8].
- **Human-in-the-Loop:** Empowers managers to supervise AI decisions, improving trust and usability [35].

- **User-Centric Design:** Tailored explanations improve trust and adoption [83].

Synthesis

Explainable AI and human-in-the-loop approaches are essential for improving trust, usability, and adoption of AI-based forecasting in institutional settings [8] [35].

Best Practices for Integrating AI, Blockchain, and IoT

- **Integration:** Real-time data acquisition (IoT), secure records (blockchain), and automation (AI) create smart, transparent, and resilient food supply chains [84] [85].
- **Hybrid Frameworks:** Address scalability, computational, and integration challenges [85].
- **Managerial Support:** Organizational factors influence adoption, requiring integrated change management [86].

Synthesis

Best practices emphasize combining IoT, blockchain, and AI for enhanced transparency, traceability, and inventory management in institutional food supply chains [84] [85] [87].

7. Conclusion

Summary and Future Outlook

AI and ML techniques, especially ensemble and hybrid models, are highly effective in predicting food demand in institutional settings when combined with IoT and big data tools. This integration not only enhances forecasting accuracy but also drives operational efficiencies by reducing food waste and optimizing inventory management. Despite these advancements, challenges such as data scarcity, model interpretability, and infrastructural limitations—particularly in low-resource settings—persist. Addressing these gaps through standardized datasets, explainable AI practices, and innovative integration of technologies like blockchain could significantly advance the practical deployment of AI in institutional food services.
Future research should focus on developing adaptive, real-time AI systems, integrated decision support platforms, and context-sensitive strategies for AI adoption, especially in low- and middle-income countries. The transformative potential of AI/ML in institutional food demand forecasting is clear, but realizing this potential requires coordinated efforts across research, industry, and policy domains [16] [17] [18].

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